

Biochemistry lab

Experiment (5)

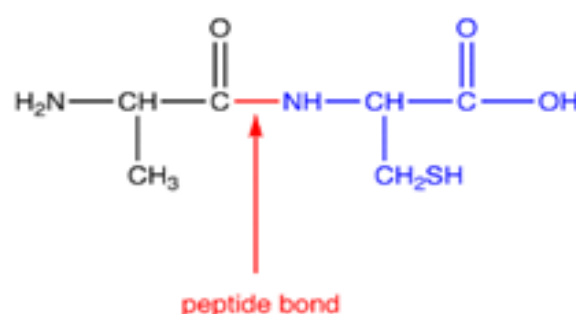


EXPIREMENT 5

GENERAL PROPERTIES OF PROTEINS.

Note: General properties of proteins differ from general properties of amino acids.

- **Peptide bonds:** are formed between two amino acids, called Amide bonds.
- **Structure of Amide bond(peptide bond):**



- Peptide bond is formed by Dehydration reaction (losing of water molecules, OH from carboxyl group and H from Amine group), that means that forming of proteins gives us water.

Don't confuse yourself! Although water is added to the outputs, it is called dehydration reaction because it dehydrates amino acids.

Dehydration reaction: adding two amino acids together, it gives water.

Hydration reaction: separating two amino acids, it consumes water.

of peptide bonds = # of H₂O molecules = # of amino acids - 1

- All proteins have peptide bonds.

In this experiment we have several tests:

First: - Biuret Test.

- This test depends on the presence of peptide bonds, since the Biuret reagent reacts with peptide bonds.
- The depth of the color obtained is a measure of number of peptide bond present in a protein
- So, we can conclude for example that Biuret test will give a deeper color with Collagen (800 a.a) than hemoglobin (141-146 a.a)
- All proteins will give positive results, but the color varies (depends on # of peptide bonds.)

NOTE: the test is not just for proteins, actually it gives positive results when it reacts with any compound containing two C=O groups linked by N or C.

"يعني, هذا التفاعل يعتبر فعال في أي مركب يحتوي على مجموعتين كربونيل مرتبطتين بنيتروجين أو كربون."

- This test gives blue-violet color. Violet is deeper than blue.
- Biuret reagent consists of: **CuSO₄(Copper Sulfate)+ NaOH (basic media)**

"يعني, هذا التفاعل يستخدم للكشف عن جميع البروتينات, لأنه يتفاعل معهم جميعا ويعطي نتائج ايجابية لهم جميعا؛ لأن جميع البروتينات تحتوي على روابط ببتيدية. لكن عمق اللون يختلف من بروتين لآخر."

● How to do this reaction?

- Take 3 test tubes.
- Add 1 ml of (5g/L) protein in each test tube [albumin, casein and gelatin]
- Add 5 drops of CuSO₄ (10 g/L)
- Add 2 ml of NaOH (10 mol/L)

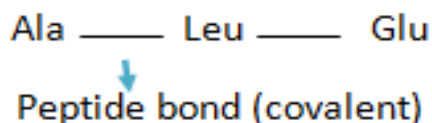
The concentration of every used solution is very important.

Otherwise, the test will not give us positive results (will be protein precipitation)

- mix through and note the color produced.
- put (+++) if the color's intensity is high.
- put (++) if the color is intermediate.
- put (+) if the color is light blue.
- there's no (-), the reaction is positive with all proteins

Second: Denaturation by heat and extreme pH

- Primary structure: is the sequence of amino acid



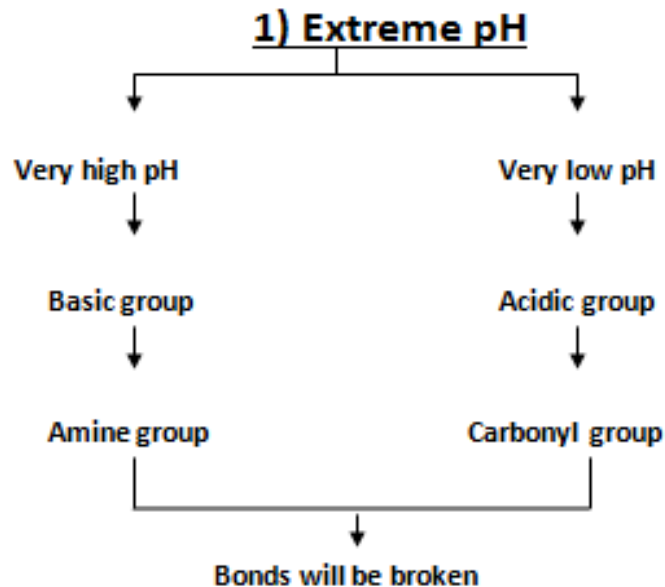
- Secondary structure: is the arrangement of backbone atoms result from H-bond (α -helix, β sheet)

tertiary structure (3D shape): the overall structure

{R groups bonds: electrostatic, hydrophobic..}

Denaturation:

Tertiary $\longrightarrow$ Secondary, Primary



- High pH will affect the amino group, and low pH will affect the carboxyl group.

Important Note: this extreme pH will break the electrostatic bonds, but it's **reversible** at room temperature .

Be careful! Not peptide bond, primary structure won't be affected by pH.

Method:

- Add 2 ml of protein in a test tube.
- Add 2ml of concentrated HNO_3 (which will reduce pH)
- You will notice a formation of yellow ring, this indicates for denaturation.

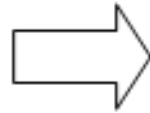
"تضعه و ال tube مائل لتظهر الحلقة بوضوح"

- **Don't forget** that the protein can be renatured after neutralizing.

2) Heat: (irreversible)

Forms of denaturation:

- 1) foam
- 2) Cloudy
- 3) ppt.



That's how we can observe denaturation.

- Heat denatures protein in an irreversible way under any condition (acidic, basic, neutral.)
- That means that it doesn't renature after getting back to neutral conditions (neutral pH).

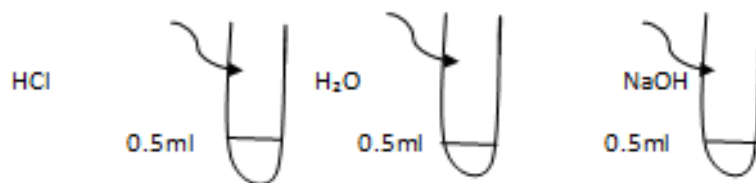
albumin in eggs is the most known example, it solidifies after heat, but it doesn't get back to liquid state even after cooling down.

"يعني، لو عرضنا البروتينات للحرارة سوف تتفكك، وهذا التفكك غير قابل للرجوع حتى لو عدلنا ال pH للحالة الطبيعية أو أعدنا الحرارة للطبيعي، البيض هو أفضل مثال."

• How to do this reaction?

- 1) Add 1ml of protein

In the 3 conditions {acidic, neutral, basic}



- 2) Put it in water bath heat (10 min)
- 3) cool back to room temperature

Third: Precipitation of proteins by heavy metals

- At slightly alkaline solutions ($\text{pH} > 7$) proteins are negatively charged (by Handerson-Hasselbalch equation) so, when we add a positive metal ion (Cu^{+2}) to alkaline proteins, it will neutralize that negative charge to form a salt, and the protein will become insoluble (Precipitate).
- After adding excess Cu^{+2} , the proteins will dissolve again, and the cloudy appearance will go.
- We don't want the solution to be too alkaline, because there will be high concentration of $[\text{OH}^-]$ and metal hydroxides will be formed, $\text{Cu}(\text{OH})_2$.

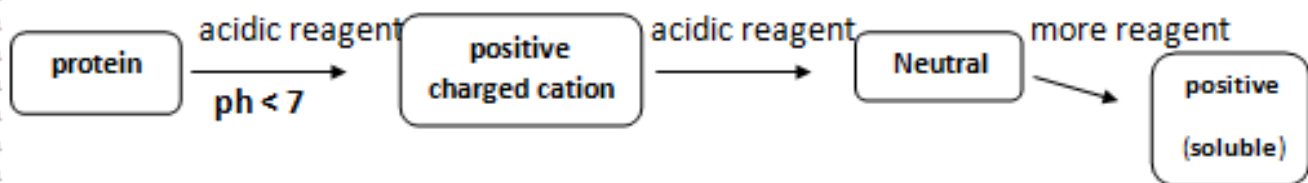
"توضيح: عند pH أعلى من 7, سيكون البروتين مشحونا بشحنة سالبة, وذائبا في الماء, فعند اضافة Cu^{+2} للبروتين سيعدل الشحنة السالبة وسيخرج البروتين من المحلول كمادة مترسبة. لكننا لا نريد محلول قاعدي أكثر من اللازم على كل حال, كي يتفاعل ال Cu^{+2} مع البروتين ليس مع OH^- "



• How to do this reaction?

- Add 2 ml of any protein to a test tube
- Add few drops of the heavy metal solution (CuSO_4)
- Notice the cloudy appearance in the tube
- Add another few drops of CuSO_4 and notice the cloudy disappearance.

Fourth: Precipitation by acidic reagent



- Here we will do the opposite, we will add an acidic reagents (negatively charged) to neutralize a positively charged protein to form an insoluble salt.
- acidic reagents: 1- sulphosalicylic acid 2- thichloroacetic acid

"عكس التجربة السابقة، سنضيف حمض مما يقلل من pH ويجعل البروتين موجبا، لكن هذا الحمض سالب الشحنة فسيعادل الشحنة الموجبة ويكون ملح لا يذوب بالماء (راسب)"

•How to do this reaction?

- Add 2 ml of any protein to test tube.
- Add few drops of any acidic reagent to the test tube.
- Notice the cloudy appearance.



Done by

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